Dear Hiring Team,

I am writing to express my interest in the Artificial Intelligence / Computer Vision Research position at Siemens Energy.

I recently completed my Master's in Computer Science from Sapienza Università di Roma, where I specialized in Machine Learning, Computer Vision, and Artificial Intelligence. My academic and research background has focused on developing and evaluating advanced machine learning systems, with particular interest in computer vision, deep learning, and continual learning methodologies.

As part of my master's thesis, I conducted extensive research on Continual Learning for Computer Vision, evaluating pretrained Vision Transformers under different learning protocols and analyzing challenges such as catastrophic forgetting. This work involved reviewing state-of-the-art scientific literature, designing experiments, implementing machine learning pipelines, and interpreting research results to identify promising directions for future development.

In addition, during my undergraduate studies I worked on computer vision projects involving CNN-based metal defect detection, gaining practical experience in image processing, feature extraction, model evaluation, and visual inspection tasks. My research experience is further supported by an IEEE-published paper focused on machine learning applications, where I analyzed multiple predictive models and identified future research opportunities.

Technically, I am proficient in Python, C++, PyTorch, TensorFlow, and modern machine learning workflows. I have experience developing experimental pipelines, conducting systematic evaluations, and translating research findings into practical implementations. My background combines strong theoretical foundations in machine learning and computer vision with hands-on experience building and evaluating AI systems.

What particularly attracts me to Siemens Energy is the opportunity to contribute to cutting-edge research that has a tangible impact on the future of sustainable energy. The prospect of working on visual intelligence, automated defect detection, asset analysis, and next-generation AI technologies within a multidisciplinary international research environment is highly motivating.

I would welcome the opportunity to discuss how my research experience, technical skills, and enthusiasm for computer vision can contribute to Siemens Energy's Artificial Intelligence organization.

Thank you for your time and consideration. I look forward to hearing from you.

Sincerely,

Swati Aggrawal